

Program overview

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Year 2023/2024
 Organization Mechanical, Maritime and Materials Engineering
 Education Bachelor Marine Technology

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor MT 2023			
1e jaar BSc MT 2023	1e Year BSc MT 2023		
Wiskunde blok			
WBMT1050	Mathematics 1	6	
WBMT1050 Toets 1	Analyse 1 - deeltentamen	3	
WBMT1050 Toets 2	Analyse 2 - deeltentamen	3	
WBMT1051	Mathematics 2	6	
WBMT1051 Toets 1	Lineaire Algebra 1 - deeltentamen	3	
WBMT1051 Toets 2	Lineaire Algebra 2 - deeltentamen	3	
Natuurkunde blok			
WB1135	Introductory Dynamics	6	
WB1530-14	Thermodynamics	6	
WB1630-20	Statics	6	
WB1631-15	Mechanics of Materials	6	
Maritieme Techniek blok			
MT1461	Introduction Maritime Technology	4	
MT1462	Geometry and Stability	4	
MT1463	Integration and skills 1	4	
MT1464	Mechanics of Maritime Structures 1	4	
MT1465	Resistance, Propulsion and Drive Systems 1	4	
MT1466	Integration and Skills 2	4	
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WBMT2048	Mathematics 3	6	
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WBMT2048 Toets 2	Differentiaalvergelijkingen - deeltentamen	3	
WBMT2049	Mathematics 4	6	
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WBMT2049 T2	Numerieke Wiskunde - deeltentamen	3	
WBMT2049 T3	Numerieke Wiskunde - practicum	0	
MT203-11 Maritieme Theorie blok	Maritieme Theorie blok		
MT2430-16	Weerstand, Voortstuwing en Aandrijving 2	6	
MT2430-16 Toets 1	WVA2 Tussentoets	1	
MT2430-16 Toets 2	Voorstuwingpracticum	0,5	
MT2430-16 Toets 3	Cavitatiepracticum	0,5	
MT2430-16 Toets 4	Dieselmotorenpracticum	0,5	
MT2430-16 Toets 5	WVA2 tentamen	3,5	
MT2432-22	Strength of Ships	6	
MT2433	Scheepsbewegingen	6	
WB2630	Advanced Mechanics	6	
WB2630 Toets 1	Rigid-Body Dynamics	3	
WB2630 Toets 2	Continuum Mechanics	3	
Maritieme Praktijk blok			
MT2429-22	Maritime Markets and Operations	6	
MT2431	Geometrie en Stabiliteit 2	6	
MT2431 Toets 1	Project Opdracht	3	
MT2431 Toets 2	G&S2 Tentamen	3	
MT204-11 Maritieme Project blok	Maritieme Project blok		
MT2434	2e Integratie Project	12	
MT2434 Toets 1	Project Opdracht(en)	10	
MT2434 Toets 2	Ethiek en Veiligheid tentamen	2	
3e jaar BSc MT 2023	3rd Year BSc MT 2023		
3e jaar BSc MT minor 2023	3rd Year BSc MT Minor 2023		
3e jaar BSc MT major 2023	3rd Year MSc MT Major 2023		
MT3420	Elektrische Systemen & Regeltechniek	6	
MT3432	Organisatie van Scheepsproductie	6	
MT3432 Toets 1	OVS Tentamen	2,5	
MT3432 Toets 2	Report	3,5	
MT3433	Complexe Belastingen en Trillingen	6	



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Bachelor MT 2023

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1e jaar BSc MT 2023

Year	2023/2024
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Wiskunde blok

WBMT1050	Mathematics 1	6
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Instructor	Dr. I.M. Smit	
Instructor	Dr. E. Emsiz	
Education Period	1 2	
Start Education	1 2	
Exam Period	1 2 3	
Course Language	Dutch	

WBMT1050 Toets 1	Analyse 1 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Instructor	Dr. I.M. Smit	
Instructor	Dr. E. Emsiz	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

WBMT1050 Toets 2	Analyse 2 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	2	
Start Education	2	
Exam Period	2 3	

WBMT1051	Mathematics 2	6
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Co-responsible for assignments	Dr. O. Jaibi	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5	
Course Language	Dutch	

WBMT1051 Toets 1	Lineaire Algebra 1 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

WBMT1051 Toets 2	Lineaire Algebra 2 - deeltentamen	3
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

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Natuurkunde blok

WB1135	Introductory Dynamics	6
Responsible Instructor	Prof.dr. P.G. Steeneken	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1530-14	Thermodynamics	6
Responsible Instructor	Prof.dr. R. Pecnik	
Instructor	Dr.ir. W.P. Breugem	
Instructor	N.F. Herting	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	3mE Department Process & Energy	

WB1630-20	Statics	6
Responsible Instructor	Prof.dr.ir. J.L. Herder	
Instructor	Dr.ing. C.H. Thill	
Instructor	Ir. D.M. Syaifoel	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	3mE Department Precision & Microsystems Engineering	

WB1631-15	Mechanics of Materials	6
Responsible Instructor	Dr.ir. M. Langelaar	
Instructor	R.F. Dooren	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Course Contents	See Dutch version	
Study Goals	See Dutch version	
Education Method	See Dutch version	
Assessment	See Dutch version	
Department	3mE Department Precision & Microsystems Engineering	

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Maritieme Techniek blok

MT1461	Introduction Maritime Technology	4
Responsible Instructor	Dr.ir. P.R. Wellens	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT1462	Geometry and Stability	4
Responsible Instructor	Dr.ing. C.H. Thill	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT1463	Integration and skills 1	4
Responsible Instructor	Dr. A.A. Kana	
Instructor	Ing. B. Goris	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	Dutch	
Course Contents	This course, MT1463 Integration & Skills 1, revolves around a main assignment. The course MT1466 Integration & Skills 2 is a continuation of this course. The main assignment consists of a design task from a fictional client, which the students must work on in groups. The groups will compete against each other for the best design. In this design competition, the students need to figure out how to approach design optimization through a variation study, incorporating the maritime knowledge acquired in quarters 1 and 2, along with the subjects of programming, computer drawing, and presentation techniques. The programming and computer drawing tutorials take place in the first quarter. The main assignment starts after these skills have been taught. In MT1463, the focus is on design, geometry, stability, ship resistance, and presentation. In MT1466, the focus shifts to ship strength, programming, teamwork, design integration, and reporting.	
Study Goals	To develop basic programming and 3D computer-aided design skills. To apply and integrate ship design knowledge to create a preliminary design (first iteration). To create a model based on the preliminary design from the first iteration. To evaluate the stability performance of various hull shape variations of the preliminary ship design. To optimize the ship design for multiple stability conditions. To conduct physical tests of a model ship hull. To communicate the key aspects of the ship design in an oral presentation.	
Education Method	The educational format for the main assignment is project-based learning, for which a few skills need to be acquired through tutorials or practical sessions. These skills include programming, technical drawing, model building, and presenting. The main assignment will mainly consist of tutorials (either in-person or online), where groups will work on the project and questions will be answered either within the groups or collectively. In addition to these tutorials, questions can also be asked on a classroom forum.	
Assessment	The students are required to give a presentation on their preliminary design. The overall grade will be determined based on subgrades for the presentation, calculated outcomes regarding stability and ship resistance, and the successful completion of practical exercises. The ratio between these subgrades and the requirements for completing the practical exercises will be defined in a grading matrix, which will be shared prior to the start of the course.	
Enrolment / Application	To participate in this course, timely registration is required. Registering for a course does not automatically mean registration for an (interim) exam, enrollment in Brightspace, or enrollment for a study program in Studielink. More information about registering for courses, the procedure, and current deadlines can be found at https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/aanmelden-vakken-en-tentamens/vakken/aanmelden-voor-vakken. To participate in interim assessments and (partial) exams, separate registration (and timely registration) for exams is required. More information about registering for exams, the procedure, and current deadlines can be found at https://www.tudelft.nl/studenten/onderwijs/vakken-en-tentamens/tentamens/aanmelden-voor-tentamens.	
Tags	*In some cases, there are entry requirements or additional criteria for admission to a course. These can be found in the OER/TER of the program: https://www.tudelft.nl/studenten/faculteiten/3me-studentenportal/onderwijs/gerelateerd/regelingen-reglementen. Design Group Dynamics/Project Organisation Integrated Modelling Optimalisation Programming Project Small groups	
Department	3mE Department Maritime & Transport Technology	
Contact	b.goris@tudelft.nl	

MT1464	Mechanics of Maritime Structures 1	4
Responsible Instructor	Dr.ir. J.H. den Besten	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT1465	Resistance, Propulsion and Drive Systems 1	4
Responsible Instructor	R.H. van Till	
Instructor	E.H.M. Ulijn	
Instructor	G.D. Weymouth	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT1466	Integration and Skills 2	4
Responsible Instructor	Dr. A.A. Kana	
Instructor	Ing. B. Goris	
Education Period	3 4	
Start Education	3	
Exam Period	4	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

Year	2023/2024
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2e jaar BSc MT 2023

Year	2023/2024
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MT201-11 Wiskunde blok

WBMT2048	Mathematics 3	6
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1 2	
Start Education	1 2	
Exam Period	Different, to be announced	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses.	
Course Contents	WBMT2048 T1 - Analysis 3: - Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss. WBMT2048 T2 - Differential Equations: - Linear first and second order equations; - Laplace transform; - Systems of linear equations - Autonomous systems and stability; - Fourier series; - Partial differential equations.	
Study Goals	Study goals for each part can be found on the corresponding StudyGuide-page (WBMT2048 Toets 1 or WBMT2048 Toets 2).	
Education Method	Lectures 4 hours per week.	
Literature and Study Materials	WBMT2048 Toets 1: Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that will also be made available via Brightspace. WBMT2048 Toets T2: William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	WBMT2048 T1: - On-campus exams. Partly digital (via Grasple), partly written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Sequences and Series) and one partial exam in week 10 covering the second part of the course (Vector Calculus). Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the entire course, which lasts 180 minutes. It is not possible to resit only one partial exam. WBMT2048 T2: - On-campus exams. Partly digital (via Grasple), partly written (see Brightspace for more details).	
Exam Hours	WBMT2048 T1: 2 partial exams of 90 minutes each. One resit of 180 minutes. WBMT2048 T2: 180 minutes.	
Permitted Materials during Tests	WBMT2048 T1: A formula sheet (see Brightspace). NO calculators. WBMT2048 T2: A table of Laplace transforms (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	See Dutch comment.	
Contact	WBMT2048@tudelft.nl	

WBMT2048 Toets 1	Analyse - deeltentamen	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Required for	(Fluid) Mechanics, Mechatronics.	
Expected prior knowledge	First year Mathematics courses: Analysis 1 and 2.	
Course Contents	- Series and sequences; - Line and surface integrals; - Integral Theorems of Green, Stokes, and Gauss.	
Study Goals	After successfully finishing this course, you are able to: - determine whether a sequence is convergent or divergent, for example by calculating its limit; - determine whether a series is convergent or divergent with a test for convergence: Integral Test, (Limit) Comparison Test, Ratio or Root Test; - determine the interval of convergence of a power series; - determine the Taylor series of a simple analytic function, for example $\sin(x)$, $\exp(x)$, $\ln(x)$, $\arctan(x)$, $(1+x)^a$; - find parametrisations of curves and surfaces (in 2D and 3D); - calculate a line integral or a surface integral of a function or a vector field; - mention the physical interpretation of these line and surface integrals; - define and formulate the physical meaning of concepts like gradient, curl, and divergence; - calculate line and surface integrals via one of the three integral theorems (Green, Stokes, and Gauss).	
Education Method	2 x 2 hour lectures per week.	
Books	Calculus, Volume 1, 2 and 3. Edwin Herman, Gilbert Strang. ISBN-13 978-1-947172-13-5, 978-1-947172-14-2, 978-1-947172-16-6 This is an open source book that also will be made available (for free) via Brightspace.	
Assessment	- On-campus exams. Partly digital (via GraspLe), partly written exams (see Brightspace for more details). - There will be two partial exams: one partial exam in week 5 covering the first part of the course (Sequences and Series) and one partial exam in week 10 covering the second part of the course (Vector Calculus). - Both exams last 90 minutes. The final grade for this course is the average of both grades for the partial exams. There is one resit, covering the complete course, which lasts 180 minutes. It is not possible to resit only one partial exam.	
Exam Hours	Two partial exams of 90 minutes each. One resit of 180 minutes.	
Permitted Materials during Tests	A formula sheet (see Brightspace). NO calculators.	
Enrolment / Application	See Dutch comment.	
Judgement	This is half of the course WBMT2048. Partial grades for T1 and T2 will be given in one decimal. The final grade is a number rounded to the nearest half. The lowest grade needed for averaging is 5,0. Example: 5,0 for T1 + 7,8 for T2 = 12,8 (Total). So the average is 6,4. The final grade will be: 6,5. The lowest Total for a valid Final Grade (a 'pass') is 11,5 (average 5,75).	
Contact	WBMT2048@tudelft.nl	

WBMT2048 Toets 2	Differentiaalvergelijkingen - deeltentamen	3
Responsible Instructor	Dr.ir. R.F. Swarttouw	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Expected prior knowledge	First year Mathematics courses: Analysis 1 and Linear Algebra 1 and 2.	
Course Contents	- First-order differential equations (linear, separable, autonomous, direction fields); - Second-order linear differential equations (Wronskian, method of reduction of order, Euler equation); - Laplace transforms and initial value problems (step function, impulse functions, the convolution integral); - Systems of first-order linear differential equations; - Non-linear autonomous systems (locally linear systems, stability, phase plane); - Partial differential equations (Heat equation, Wave equation, Laplace's equation); - Method of separation of variables; - Boundary value problems; - Fourier series (even and odd functions).	
Study Goals	After successfully finishing this course, you are able to: - distinguish between ordinary and partial differential equations; - distinguish between linear and nonlinear differential equations; - solve linear and separable first-order differential equations; - solve autonomous first-order differential equations and determine the behavior near equilibrium solutions; - solve homogeneous linear differential equations with constant coefficients; - solve the Euler equation; - determine particular solutions of non-homogeneous linear differential equations with the method of undetermined coefficients; - use Laplace transforms to solve initial value problems of linear differential equations with constant coefficients; - solve systems of homogeneous linear first-order differential equations with constant coefficients; - determine particular solutions of systems of non-homogeneous linear first-order differential equations with the method of undetermined coefficients; - determine particular solutions of systems of non-homogeneous linear first-order differential equations with the method of variation of parameters; - determine the behavior of solutions near equilibrium points of autonomous systems of differential equations; - determine the Fourier (cosine- and/or sine-)series of a function; - solve the heat or diffusion equation with the method of separations of variables; - solve the wave equation with the method of separations of variables; - solve the Laplace equation with the method of separations of variables.	
Education Method	2 x 2 hour lectures per week.	
Books	William E. Boyce, Richard C. DiPrima & Douglas B. Meade: Elementary Differential Equations and Boundary Value Problems. International Adaption, Twelfth Edition, Wiley, 2022. ISBN 978-1-119-82051-2.	
Assessment	On-campus exams. Partly digital (via Graspale), partly written exams (see Brightspace for more details). The exam lasts 180 minutes.	
Permitted Materials during Tests	A table with Laplace transforms (see Brightspace). Bring your own printed version to the exam!	
Judgement	This is half of the course WBMT2048. Partial grades for T1 and T2 will be given in one decimal. The final grade is a number rounded to the nearest half. The lowest grade needed for averaging is 5,0. Example: 5,0 for T1 + 7,8 for T2 = 12,8 (Total). So the average is 6,4. The final grade will be: 6,5. The lowest Total for a valid Final Grade (a 'pass') is 11,5 (average 5,75).	
Contact	WBMT2048@tudelft.nl	

WBMT2049	Mathematics 4	6
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Instructor	Dr. N.V. Budko	
Education Period	3 4	
Start Education	3 4	
Exam Period	3 4 5 Different, to be announced	
Course Language	Dutch	

WBMT2049 T1	Kansrekening en Statistiek - deeltentamen	3
Responsible Instructor	Dr.ir. R. Versendaal	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

WBMT2049 T2	<i>Numerieke Wiskunde - deeltentamen</i>	3
Responsible Instructor	Dr. N.V. Budko	
Instructor	Drs. A.T. Hensbergen	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

WBMT2049 T3	<i>Numerieke Wiskunde - practicum</i>	0
Responsible Instructor	Ir. D. den Ouden-van der Horst	
Education Period	4	
Start Education	4	
Exam Period	Different, to be announced	
Course Language	Dutch	

Year	2023/2024
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MT203-11 Maritieme Theorie blok
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MT2430-16	Weerstand, Voortstuwing en Aandrijving 2	6
Responsible Instructor	Dr.ir. P. de Vos	
Instructor	Prof.dr.ir. T.J.C. van Terwisga	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Study Goals	After successfully completing this course the student is able to: With respect to: - Resistance: - Describe ITTC 72 breakdown of ship resistance into components - Explain (and apply some) methods for resistance prediction: model tests, systematic series, empirical prediction programs, RANS - Propulsion: - Describe and apply actuator disk theory and Open Water diagrams - Explain lifting line theory and propeller design - Explain the relation between propeller geometry and performance and cavitation - Explain cavitation, erosion and pressure fluctuations - Calculate Thrust, Torque, Power, Efficiency, etc. of propellers - Resistance and Propulsion (R&P): - Derive and explain properties of fluid dynamic models in R&P: RANS, Euler, Laplace, thin boundary layer equations - Describe Propeller-hull interaction - Explain application of BEM (panel methods) and RANS methods in R&P - Describe propeller-hull optimization in ship design and Energy Saving Devices - Driving components (prime movers like diesel engine or gas turbine): - Explain and apply torque-speed/power-speed characteristics, construction types, ideal process models and actual processes, turbo-charging concept - Calculate Torque, Power, Efficiency, Mean Effective Pressure, fuel and air consumption, air excess ratio and load related units (in thermodynamical terms) of engines - Propulsion systems (combination of drive + propulsion components to overcome resistance): - Explain different lay-outs of propulsion systems for different ships (applications) - Match propulsor to ship and prime mover to propulsor - Analyse (off-)design behaviour of propulsion systems - Pump pipe flow systems: - Explain different lay-outs of pump pipe flow systems for different applications (in ships) - Analyse stationary behaviour of pump pipe flow systems in Q-H diagram - Calculate pressures, (flow) velocities, power, etc. - Emissions: - Describe different emissions from ships (to air and water) - Explain the root of harmful emissions from chemical composition / processes - Describe regulations to limit emissions from ships and their effects - Describe possible counter measures / emission abatement systems Course material underlying theory: - Conservation laws from physics (conservation of mass, momentum and energy). - Fluid mechanics (describing boundary layer, continuity, properties of material flow (liquid), Bernoullis equation). - Thermodynamics (ideal gas, cyclic processes, properties of material flow (gaseous), second law). - Chemistry (high school chemistry is sufficient)	
Department	3mE Department Maritime & Transport Technology	

MT2430-16 Toets 1	WVA2 Tussentoets	1
MT2430-16 Toets 2	Voorstuwingspracticum	.5
MT2430-16 Toets 3	Cavitatiepracticum	.5
MT2430-16 Toets 4	Dieselmotorenpracticum	.5
MT2430-16 Toets 5	WVA2 tentamen	3.5

MT2432-22	Strength of Ships	6
Responsible Instructor	Dr. C.L. Walters	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Description box (5 ECTS): The course consists of four parts: - Strength of ships: (1) lectures + (2) laboratory exercise - Material science: (3) lectures + (4) laboratory exercise (1) Strength of ships lectures Special topics related to the limit and ultimate strength of ships will be introduced, including buckling (beam and plate), plasticity, fatigue, and fracture.: a) Stress concentration factors b) Buckling of columns: imperfections, eccentricities, Euler, Johnson, Perry-Robertson; c) Buckling of plates: loading, boundary conditions, geometries; d) Effective width of plates: stiffness differences, shear deformations, non-symmetrical/curved cross-sectional forms, post-buckled stress distribution and types of loading; e) Plasticity: form factors, maximum capacity of beams, hinge line theory f) Fracture and fatigue: S-N curves, fatigue crack growth rate, Charpy testing, linear elastic fracture theory, elastic plastic fracture theory. The lecture materials consist of presentations (PowerPoint). The following two reference books are recommended: - Fracture Mechanics: Fundamentals and Applications, ISBN 9781498728133 - Ship Structural Analysis and Design, ISBN 978-0-939773-78-3 (available for download from TU Delft library) (2) Strength of ships laboratory exercise: plate test (3) Material sciences lectures a) Strengths of materials: characteristics of steel and some non-ferrous ((mechanical) materials, density, structure and properties). b) Strengthening mechanisms on the microstructural scale. c) Thermomechanical processes of metals (deformation, recrystallization, phase transformations). d) Tension test: elastic modulus, yield strength, ultimate tensile strength, failure strain, necking, hardening, and plastic instability. e) Most common welding processes, process descriptions, advantages and disadvantages. f) Influence of welding on materials microstructure, residual stress, deformations, common defects. (4) Material sciences laboratory exercise	
Study Goals	(1) + (2) Strength of ships The recognition of various structural elements (beam and plate elements) of ships with associated supporting conditions and loads. The ability to dimension these structural elements on the basis of strength, stiffness and structural stability. The determination of the stress condition and maximal strength capacity for plate elements loaded in the plane, based on measurements (via digital image correlation). (3) + (4) Materials sciences Insight into the relationship between structure and properties of various materials and into the influence of the production processes on the structure of materials and the use of these materials sciences insights into the design of structures. The ability to describe basic aspects of (fracture-) mechanical material behavior, including the determination of this behavior and the ability to make a simple failure calculation, e.g. for a welded connection. The development of basic knowledge of selected welding processes. Insight into the influence of welding on local microstructures and properties around a welded and the influence of welding on structures. Obtaining insight into the basic principles of melt welding process. Awareness of mechanical tests and the interpretation of results of a tension test.	
Education Method	Lectures and laboratory exercises	
Assessment	a) Laboratory exercise for strength of ships: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. b) Laboratory exercise for material science: creation of a report, graded on a pass/fail basis. Students must pass this for participation in the written exam. This is otherwise not accounted for in the determination of the grades. 0.5 ECTS. c) Written exam: this exam consists of a combination of the strength of ships and materials science. 5 ECTS	
Enrolment / Application	See Dutch info	
Department	3mE Department Maritime & Transport Technology	
Judgement	For the regulations (e.g. signing up for education and validity of grades) please refer to the Educational regulations, http://onderwijs.3me.tudelft.nl/reglementen	

MT2433	Scheepsbewegingen	6
Responsible Instructor	Dr.ir. H.J. de Koning Gans	
Instructor	Dr.ir. H.J. de Koning Gans	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	The course lectures the basic knowledge and theory concerning shipmotions in waves. The course addresses: * Wave theory in deep and shallow water * Calculation methods based on potential flow theory, like 2D strip-theory * Equations of motions for a ship in waves * Lifting theory applicable to rudders and fins * Experimental techniques to establish ship motions (the towing tank experiments)	
Study Goals	After the course the student is able to: * Calculate pressures and velocities in regular waves * Calculate local motions and accelerations on board a vessel * Explain the concept of transfer functions for vessel motions * Explain the forces on lifting surfaces like rudders As well, the student is able to: * Explain the motion equations for a ship and explain the theory of ship motions * Describe the parameters in the motion equations and how they can be established by means of experimental techniques	
Education Method	Lectures and Workshops	
Literature and Study Materials	Lecture material will be posted on Brightspace	
Assessment	Written examen	
Remarks	It is essential that the following course have been followed and that the material is 'known' at the start of the lectures. This concerns: "Scheepshydrostatica, Analyse, Lineaire algebra ,Differentialvergelijkingen, Rigid body dynamics and Mechanics".	
Department	3mE Department Maritime & Transport Technology	

WB2630	Advanced Mechanics	6
Responsible Instructor	Dr.ir. L.F.P. Noel	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Course Contents	See the respective parts T1 and T2	
Study Goals	See the respective parts T1 and T2	
Education Method	See the respective parts T1 and T2	
Assessment	See details in T1 and T2	
Enrolment / Application	see Dutch version	
Department	3mE Department Biomechanical Engineering 3mE Department Precision & Microsystems Engineering	
Judgement	see Dutch version	

WB2630 Toets 1		Rigid-Body Dynamics	3
Responsible Instructor	Dr.ir. A.H.A. Stienen		
Instructor	W. Wolfslag		
Contact Hours / Week x/x/x/x	lectures 2/0/0/0 Practicals 2/0/0/0 Office hour 2/0/0/0		
Education Period	1		
Start Education	1		
Exam Period	1 2		
Course Language	Dutch		
Expected prior knowledge	Linear Algebra, Dynamics (2D)		
Course Contents	1) Kinematics of a rigid body in 3D - Description of single and successive 3D rotations via Euler angles, rotation matrices, axis and angle of rotation. - Angular velocity and instantaneous axis of rotation - Transport Theorem 2) Kinetics of a rigid body in 3D - The inertia tensor - Principle of work & energy - Newton-Euler equations - Free rotational motion, gyroscopic effects - Numerical integration 3) Systems with n degrees of freedom - Generalized coordinates - Principle of virtual work - Lagranges equations 4) Methods to check plausibility of generated results, such as - Units notation and checks - Numerical approximation - Checking validity of assumptions - Treatment of special cases		
Study Goals	Kinematics of a rigid body in 3D The student is able to describe rotations of a rigid body in 3D via Euler angles, rotation matrices, and the principal axis of rotation. transform one description of rotations into the other, and re-write successive rotations as a single rotation in the different representations. describe angular velocity of a rigid body in 3D and find the instantaneous axis of rotation apply transport theorem to take derivatives of vectors in rotating frames Kinetics of a rigid body in 3D The student is able to calculate the inertia tensor for a rigid body via integration analyze the inertia tensor and find the principal axes derive the equations of motion for a rigid body in 3D using the principle of work and energy or using the Newton-Euler equations recognize and explain gyroscopic effects Systems with n degrees of freedom The student is able to formulate kinetic and potential energy for systems of particles and for rigid bodies in generalized coordinates apply Lagranges methods to simple systems of particles and to rigid bodies to obtain the equations of motion recognize conservative systems Generating reliable results to mechanical problems The student is able to test the plausibility of calculation steps and final results be aware of and check validity of assumptions whenever applying learned concepts		
Education Method	Each week, interactive on-campus lectures and tutored sessions are organized. Students study the relevant course material before each lecture. For this, they can use the information and material provided on Brightspace and the recommended books. During the tutored sessions, students can work on homework exercises and ask for further explanation. In the first week, a formative digital test is provided to help students understand their current level with respect to prerequisites; this test does not count towards the courses grade.		
Books	Vallery/Schwab "Advanced Dynamics", in 1st (black), 2nd (blue), or 3rd (red) edition		
Assessment	The assessment of this course is a single on-campus exam, taken individually, which will result in the final grade. The exam is taken digitally, in computer rooms, during the exam period following the course. The resit of the course is in the same format later in the year. The final exam are organized on-campus. They are proctored and closed-book, with students provided a reference sheet that was also used throughout the course. For information on how the grades of WB2630-T1 (Rigid Body Dynamics) and WB2630-T2 (Continuum Mechanics) are combined in the final grade for WB2630 (Advanced Mechanics), please see the Brightspace page of WB2630.		
Enrolment / Application	See Dutch version		
Department	3mE Department Biomechanical Engineering		
Judgement	See Dutch version		
Contact	Rigid-Body-Dynamics-3me@tudelft.nl. Please do not use lecturers' individual e-mail addresses.		

WB2630 Toets 2	Continuum Mechanics	3
Responsible Instructor	Dr.ir. L.F.P. Noel	
Contact Hours / Week x/x/x/x	lectures 2/0/0/0 Practicals 2/0/0/0 Office hourse 2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Linear 3D continuum theory 1. Kinematics equations in 3D 2. Deformation tensor in 3D, linear and nonlinear 3. Stretch in arbitrary direction 4. Equilibrium equations in 3D 5. Stress vector and tensor in 3D, linear 6. Principal stresses and directions, normal and shear stress acting on a plane 7. Constitutive equations in 3D 8. Generalized Hooke's law 9. Plane stress/strain state Energy principles 1. Principle of virtual work (PVW) without/with deformations 2. Principle of minimum potential energy (MPE) 3. Approximate solutions based on MPE Discrete linear systems 1. Systematic way of solving problems with n degrees of freedom (DOF) 2. Analysis of discrete linear systems (spring, truss) DOF, generalized deformations, generalized stresses, generalized constitutive equations, discrete continuity and equilibrium, member/global stiffness, assembly, boundary conditions, solution Finite Element Method 1. Approximations based on MPE or the PVW 2. Derivation of simple elements and isoparametric elements 3. Analysis of structures with the FEM DOF, element/global stiffness, assembly, boundary conditions, distributed loads, solution 4. Discussion on element choice, mesh refinement, convergence	
Study Goals	By the end of the course, the student is able to: Linear 3D continuum theory 1. Recognize situations where the continuum theory is applicable 2. Derive kinematics equations in 3D a. Derive relation between displacements and deformations b. Provide interpretation for the components of the strain tensor c. Recognize situations where kinematics equations are applicable 3. Use kinematics equations a. Determine deformations, a strain tensor for known displacements b. Determine deformations, stretches in specified directions c. Explain the relevance of and the procedure to measure deformations in specified directions 4. Derive the equilibrium equations in 3D a. Explain equilibrium equations in 3D b. Explain the relations between external forces, internal forces, and stresses c. Recognize situations where equilibrium equations are applicable 5. Use equilibrium equations a. Determine a stress vector, a stress tensor b. Determine a normal stress, a shear stress c. Determine principal stresses and directions 6. Derive generalized Hooke's law in 3D a. Describe its relevance and limitations 7. Apply generalized Hooke's law 8. Identify plane stress or plane strain states Energy principles 1. Describe the principle of virtual work (PVW) 2. Derive equilibrium equations using the PVW 3. Recognize situations where the PVW is applicable 4. Describe the principle of minimum potential energy (MPE) 5. Derive equilibrium equations using the MPE 6. Recognize situations where the MPE is applicable 7. Build approximate solutions using the MPE Discrete linear systems 1. Build discrete linear systems for given structures 2. Analyze structures with discrete linear systems a. Build local member stiffness matrices b. Build global system stiffness matrices c. Set up system of equations for discrete linear systems 3. Solve problems using discrete linear systems a. Impose boundary conditions b. Solve for displacements at joints c. Post-process for reaction forces, stresses, and strains Finite Element Method 1. Describe the basic concepts of the Finite Element Method (FEM) 2. Derive simple and isoparametric elements based on the PVW or on the MPE a. Build a geometry and a displacement approximation over an element b. Build the stiffness matrix for an element c. Build energy conjugated loads for an element d. Perform numerical integration over an element 3. Analyze structures with the FEM a. Build local element stiffness matrices b. Build global structure stiffness matrices	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Marine Technology

Maritieme Praktijk blok

MT2429-22	Maritime Markets and Operations	6
Responsible Instructor	Dr. A.A. Kana	
Instructor	Prof.dr.ir. E.B.H.J. van Hassel	
Instructor	Ir. N. de Vries	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT2431	Geometrie en Stabiliteit 2	6
Responsible Instructor	Dr.ing. C.H. Thill	
Instructor	Ing. B. Goris	
Assistent	J.G. den Ouden	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	
Course Contents	Please refer to Dutch page	
Department	3mE Department Maritime & Transport Technology	

MT2431 Toets 1	Project Opdracht	3
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MT2431 Toets 2	G&S2 Tentamen	3
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Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Marine Technology

MT204-11 Maritieme Project blok

MT2434	2e Integratie Project	12
Responsible Instructor	Dr. A.A. Kana	
Instructor	E.H.M. Ulijn	
Instructor	Dr. P.E. Vermaas	
Instructor	Ing. B. Goris	
Instructor	Dr. E. Papadimitriou	
Instructor	Ir. N.C.H. Troost	
Instructor	Ir. N. de Vries	
Education Period	3	
	4	
Start Education	3	
Exam Period	4	
Course Language	Dutch	
Tags	Analysis Design Ethics Integrated Mechanics Modelling Optimalisation Process Project Small groups Sustainability	
Department	3mE Department Maritime & Transport Technology	

MT2434 Toets 1	Project Opdracht(en)	10
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MT2434 Toets 2	Ethiek en Veiligheid tentamen	2
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	Onderdeel 1 - Veiligheidskunde Het onderdeel veiligheidskunde is erop gericht een bewustzijn op te bouwen van de risico's in de dagelijkse praktijk en inzicht te geven in instrumenten om die risico's te identificeren en te beheersen. Als introductie tot de veiligheidskunde wordt ingegaan op: Bewustwording van gevaren en risico's, veiligheid als gevolg van menselijke, technische en organisatorische factoren, Methoden voor het systematisch analyseren van risico's. Aansluitend worden verschillende cases uitgewerkt, die in combinatie met Ethiek worden voorgelegd. Vanuit die lijn zijn de leerdoelen geformuleerd die, met op de discipline afgestemde cases, worden toegelicht. Onderdeel 2 - Ethiek Het ethiek onderdeel richt zich op de vaardigheden die nodig zijn om ethische en maatschappelijke problemen, waarmee ingenieurs in hun beroepspraktijk te maken kunnen krijgen, te herkennen en er verantwoordelijk mee om te gaan. Als introductie tot de Ethiek wordt ingegaan op: Het verwerven van (achtergrond) kennis en (cognitieve) vaardigheden die noodzakelijk zijn voor het herkennen, analyseren, en oplossingsgericht redeneren over, en bediscussiëren van ethische en maatschappelijke problemen waarmee ingenieurs in de beroepspraktijk te maken krijgen. Het in staat zijn tot zelfstandige gefundeerde oordeelsvorming over deze problemen en over mogelijke of voorgestelde oplossingsrichtingen. Het bevorderen van maatschappelijk verantwoordelijkheidsbesef en voorbereiding op een maatschappelijk verantwoorde beroepsuitoefening. Aansluitend hierop moet een uitgebreide case worden uitgewerkt, die in combinatie met Veiligheidskunde wordt voorgelegd. Vanuit deze insteek zijn de leerdoelen geformuleerd, die met op de discipline afgestemde cases worden toegelicht.	
Study Goals	De student kan: (Veiligheidskunde) 1) de gangbare definities van gevaar en risico reproduceren en hun betekenis voor ontwerp en gebruik van technische systemen illustreren 2) enkele belangrijke veiligheidsprincipes en -normen weergeven 3) onderkennen dat gevaren en risico's rond bestaande of nieuwe ontwerpen of systemen kunnen worden geïdentificeerd door gebruik te maken van enkele veelgebruikte analyse- of model technieken zoals het H-B-T (Hazard- Barrier-Target) model, de Veilige Envelop etc. 4) de interactie tussen mens en machine als bron van falen herkennen 5) risico's modelleren met decompositie-technieken (Foutenboom, Gebeurtenissenboom, het BowTie model) die ook ten grondslag liggen aan Kwantitatieve Risico Analyse 6) de problemen onderscheiden die verbonden zijn aan het afleiden van kwantitatieve gegevens (Ethiek) 7) de verschillende manieren herkennen waarop begrippen als verantwoordelijkheid en morele of maatschappelijke verantwoordelijkheid in het taalgebruik gehanteerd worden, en de verschillende opvattingen over de verantwoordelijkheid van ingenieurs 8) de mogelijke functies en beperkingen van beroepsregels verwoorden 9) de barrières herkennen die ingenieurs veelal ondervinden in de beroepspraktijk als ze verantwoord willen handelen 10) reflecteren op het vergroten van de ethische aanvaardbaarheid van risico's door ontwerpaanpassingen 11) argumenten formuleren en verdedigen ten aanzien van verantwoordelijkheid en aansprakelijkheid van actoren	
Education Method	werkcolleges en project: deelname is verplicht	
Assessment	De eindtoets bestaat uit een multiple choice tentamen. Indien er sprake is van onvoorziene omstandigheden, of maatregelen als gevolg van COVID-19, wordt als volgt afgeweken van de voorgescreven toetsing: Remote opdracht. Het cijfer voor deze cursus is gebaseerd op de resultaten van huiswerkopdrachten voor Ethiek, respectievelijk voor Veiligheidskunde en op het tentamen. Een minimum score van 5.0 is vereist voor elke opdracht. Het eindcijfer is gebaseerd op de volgende indeling: 25% VK-opdrachten, 25% Ethiek-opdrachten en 50% tentamen.	
Department	3mE Department Maritime & Transport Technology	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Marine Technology

3e jaar BSc MT 2023

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Marine Technology

3e jaar BSc MT minor 2023

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Bachelor Marine Technology

3e jaar BSc MT major 2023

MT3420	Elektrische Systemen & Regeltechniek	6
Responsible Instructor	Dr.ir. H. Polinder	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Required for	MSc Maritime Technology	
Course Contents	- Introduction to drive and energy systems - Electricity and electrical circuits - Magnetism and magnetic circuits - Power electronics - DC machines - AC voltages, currents, powers and phasors - Transformers - Synchronous machines (including PM machines) - Induction machines - Choices in electrical drive and energy systems - Introduction to control - Dynamic models and transfer functions - Dynamic response - Analysis of feedback control - Frequency response	
Study Goals	- Calculate DC and AC (single phase and three-phase) voltages and currents in simple electrical circuits (such as equivalent circuits of electrical machines) with sources, resistive, capacitive and inductive loads using Kirchhoffs laws. - Sketch phasor diagrams with voltage and current phasors. - Calculate active, reactive and apparent power in AC circuits. - Calculate energy stored in capacitors and inductors. - Describe the most important power semiconductors (diodes, IGBTs, MOSFETs) and their switching behaviour. - Describe how power electronic converters use these switches to convert DC from one voltage level to another (choppers or phase legs using IGBTs or MOSFETs), AC into DC (passive rectifiers using diodes or active rectifiers using IGBTs or MOSFETs) or DC into AC (phase legs using IGBTs or MOSFETs). - Derive an expression for the output voltage of a single-phase or three-phase diode bridge rectifier as a function of input voltage in continuous conduction mode (neglecting commutation). - Derive an expression for the output voltage of a DC-DC converter as a function of duty cycle and input voltage in continuous conduction mode. - Calculate induced voltage using Faradays law. - Calculate electromagnetic forces and torques using Lorentz law or the power balance. - Derive voltage equations and equivalent circuits of transformers, DC machines, synchronous machines, induction machines and permanent magnet machines. - Derive performance characteristics based on these equivalent circuits (torque-speed characteristics). - Describe advantages and disadvantages of electrification of drives. - Select suitable machine and drives for different applications. - Recognize open-loop and closed-loop control systems. - Describe elements in the control loop. - Describe control systems using block diagrams. - Model and linearize simple physical systems (in particular mechanical, electrical, electromechanical, fluid and thermodynamic systems). - Derive transfer functions from simple linear differential equations, using Laplace transforms. - Analyse the dynamic behaviour of closed-loop systems based on transfer functions in terms of step and frequency response or of poles and zeros. - Set up PID controllers for simple systems and apply fine-tuning rules (such as Ziegler & Nichols). - Interpret Bode diagrams for 1st and 2nd order systems. - Apply the aforementioned control methods to electrical machines by modelling an electrical machine and drawing up a transfer function for it.	
Education Method	Lectures	
Books	T. Wildi, "Electrical Machines, Drives and Power Systems" G.F. Franklin, J.D. Powell, A. Emami-Naeini, Feedback Control of Dynamic Systems	
Assessment	The examination will be as follows: written exam, closed book + obligatory practical. Students pass the course if the result of the exam is $\geq 5,8$ and students passed the practical, and is equal to the the result of the exam. The result of the practical is only valid in the year that the practical is done. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: written exam, open book, online.	
Department	3mE Department Maritime & Transport Technology	
Contact	h.polinder@tudelft.nl	

MT3432	Organisatie van Scheepsproductie	6
Responsible Instructor	Dr.ir. J.F.J. Pruijn	
Responsible Instructor	Ir. M. van der Schaaf	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Department	3mE Department Maritime & Transport Technology	

MT3432 Toets 1	OVS Tentamen	2.5
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MT3432 Toets 2	Report	3.5
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MT3433	Complexe Belastingen en Trillingen	6
Responsible Instructor	Dr. A. Grammatikopoulos	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	This course requires mastery of the following subjects: Matrix algebra (WBMT1051: Wiskunde2) Eigenvalues and Eigenvectors (WBMT1051: Wiskunde 2) Partial differential equations (WBMT2048: Wiskunde 3) Newton's Laws (WB1135: Introductory Dynamics) Dynamics of rigid bodies (WB2630: Advanced Mechanics) Tension, bending, torsion, elongation, shear (WB1631-15: Sterkteleer) Complex numbers, trigonometric functions, logarithms Practical use of Python (alternatively MATLAB or Maple, but support will only be provided for Python)	
Course Contents	1) General knowledge: Phasor notation Fourier transform (frequency content of time signals) Fluid effects on vibration (added mass/damping/stiffness) 2) Vibration of discrete systems with 1 or more degrees of freedom: Equations of Motion (free body diagram, Lagrange method) Free response (natural frequencies, mode shapes, damping) Forced response (transfer functions) 3) Vibration of continuous systems: Analytical methods (natural frequencies, mode shapes) Approximation methods System coupling 4) Structural waves and noise: Mechanical waves Noise fields and sound paths Transmission, reflection, absorption, radiation Decibel scale and frequency bands Noise propagation on ships 5) Application: Sea-induced vibration Machinery-induced and propeller-induced vibration Countermeasures Experiments & measurements	
Study Goals	The student can predict, measure and assess the structural dynamic behaviour (noise and vibration levels) of complex discrete and continuous mechanical systems. Specifically, this means that the student: Can determine the dynamic behaviour of 1DOF systems, 2DOF systems and simple continuous systems analytically. Can use numerical methods within a programming environment, such as Python, to solve more complex systems, for which an analytical solution does not exist. Can apply the modal summation principle to MDOF systems and continuous systems. Can describe the main sources, transmission paths and analysis methods for noise and vibration. Can suggest countermeasures to reduce noise and vibration.	
Education Method	The material will be delivered through lectures and tutorials. In the event of unforeseen circumstances or measures as a result of COVID-19, these lectures and tutorials may take place online. In order to process the lecture material and to prepare for the exam, the student is expected to individually complete exercises every week. Some of the exercises will involve development of code during the tutorials and others will be in the form of online quizzes. The code developed within the tutorials will be used to perform the practicum calculations. The aspects of analytical and numerical prediction, measurement and analysis of the dynamic behaviour of a mechanical system will be treated in a practicum in small groups. In the event of unforeseen circumstances or measures as a result of COVID-19, it is possible that the measurements of the practicum may be cancelled. In that case, only the analytical and numerical calculations will be performed and reported. The total expected time needed to complete the course is 168 hours per student, with the following breakdown: 32 hours of lectures (4 hours per week) 16 hours of tutorials (2 hours per week) 16 hours for weekly practice quizzes (2 hours per week) 29 hours for the practicum (5 hours for initial planning and analytical calculations, 15 hours for numerical calculations, 3 hours of measurements, 9 hours for analysis and reporting) 32 hours for self-study (4 hours per week during the semester, can be partly used to further support other aspects, such as the weekly quizzes or the practicum) 40 hours of preparation for the exam 3 hours exam	
Reader	The MT3402 reader Scheepstrillingen en geluid is used for this course.	
Assessment	The assessment takes place in 2 parts: Exam (85%) Assignments (15%) If possible, the examination will be held physically in the form of a written examination. In the event of unforeseen circumstances or measures due to COVID-19, the assessment will take place online using MapleTA/Mobius.	
Permitted Materials during Tests	The following materials are allowed during the exam: Calculator (non-programmable) Ruler or triangle Blue or black pen Any material in hard copy (printed or handwritten)	
Remarks	You must register to participate in this course. Registration for a course is not the same as registration for a (partial) exam, registration in Brightspace or registration for a study in Studielink.	
Department	3mE Department Maritime & Transport Technology	

MT3BEP	Bachelor End Project	12
Responsible Instructor	Prof.dr.ir. B.J. Boersma	
Instructor	Dr.ir. H.J. de Koning Gans	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	
Course Contents	Het Bachelor Eind Project (BEP) is het laatste onderdeel van de bachelor Maritieme techniek (MT), heeft een omvang van 12 ECTS en wordt uitgevoerd in periode 1 en 2 of periode 3 en 4. Het project fungeert als capstone, een afsluitend en samenvattend project voor de gehele bachelor. Het project wordt uitgevoerd in groepen van 4 studenten. De opdracht van het project is voor iedere groep studenten verschillend maar heeft in alle gevallen betrekking op een multidisciplinaire (ontwerp)opdracht waarin een duidelijke onderzoekscomponent aanwezig is. Deze (ontwerp)opdracht kan betrekking hebben op een heel schip, een deel van een schip of een aan scheepsbouw of scheepvaart gerelateerd proces. Opdrachten worden geformuleerd door of namens U(H)Ds en hoogleraren van MT. De opdracht bevat zowel een verdiepende onderzoekscomponent als een multidisciplinaire ontwerp en/of analyse component, maar de balans tussen beiden kan per opdracht binnen redelijke grenzen vrij gekozen worden. Aan beide componenten wordt tenminste ruwweg 25% van de tijd besteed. De aard van de opdracht is altijd dusdanig dat het de studenten noodzaakt om tenminste 2 disciplines binnen MT te combineren.	
Department	3mE Department Maritime & Transport Technology	

Dr.ir. J.H. den Besten

Unit	Mech, Maritime & Materials Eng
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Dr.ir. W.P. Breugem

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Dr. N.V. Budko

Unit	Elektrotechn., Wisk. & Inform.
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Department	Precision and Microsystems Engineering
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Dr. A. Grammatikopoulos

Department	Ship and Offshore Structures
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Prof.dr.ir. E.B.H.J. van Hassel

Department	Ship Design, Production and Operations
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Department	Precision and Microsystems Engineering
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N.F. Herting

Department	Energy Technology
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J.G. den Ouden

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Ir. D. den Ouden-van der Horst

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Prof.dr. R. Pecnik

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Dr.ir. J.F.J. Pruijn

Unit	Mech, Maritime & Materials Eng
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Ir. M. van der Schaaf

Unit	Mech, Maritime & Materials Eng
Department	Ship Design, Production and Operations

Dr. I.M. Smit

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Prof.dr. P.G. Steeneken

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Unit	Mech, Maritime & Materials Eng
Department	Dynamics of Micro and Nano Systems
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Unit	Technische Natuurwetenschappen
Department	QN/Steeneken Lab
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Dr.ir. A.H.A. Stienen

Department	Biomechatronics & Human-Machine Control
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Dr.ir. R.F. Swarttouw

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Ir. D.M. Syaifoel

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Prof.dr.ir. T.J.C. van Terwisga

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Department	Ship Hydromechanics and Structures
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Dr.ing. C.H. Thill

Unit	Mech, Maritime & Materials Eng
Department	Ship Design, Production and Operations
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